GARLIC - Genomic Autozygosity Regions Likelihood-based Inference and Classification v2.0.0 User Manual

Zachary A Szpiech

October 22, 2026

Contents

1 Introduction **3**

2 Obtaining GARLIC **3**

3 Basic Usage **3**

4 Statistics implemented **5**

5 Program Options **6**

5.1 Input Files . . . 7
5.1.1 VCF input . . . 7
5.1.2 Cohorts of mixed depth: `--stratify-cutoff` . . . 8
5.1.3 Reference and mapping bias: `--exclude-regions` . . . 8
5.1.4 File formats . . . 9
5.1.5 `--tped` . . . 10
5.1.6 `--tped-missing` . . . 10
5.1.7 `--tfam` . . . 10
5.1.8 `--tgls` . . . 10
5.1.9 `--centromere` . . . 10
5.1.10 `--freq-file` . . . 10
5.1.11 `--map` . . . 10
5.2 Output Files . . . 11
5.2.1 `--out` . . . 13
5.3 Multiple populations . . . 13
5.3.1 Where the labels come from . . . 13
5.3.2 Output files . . . 13
5.3.3 `--pool-populations` . . . 14
5.3.4 When a population cannot be analysed . . . 14
5.4 Sex chromosomes . . . 14
5.4.1 Declaring the system . . . 14
5.4.2 What the analysis does . . . 14
5.4.3 Pseudoautosomal regions . . . 15
5.4.4 Chromosome names . . . 16
5.5 Counting classified genotypes . . . 16
5.5.1 Counting against calls that already exist . . . 17
5.6 Controlling GARLIC . . . 17

6 Change Log **17**

A Command line reference **21**

1 Introduction

Extended tracts of homozygosity in individual genomes manifest as a result of haplotypes inherited identical by descent (IBD) from both biological parents. These runs of homozygosity (ROH) are important genomic features, with their length distributions and genomic locations informative about population history and useful for the mapping of recessive loci contributing to Mendelian and complex diseases. However, precisely because the length distribution of ROH is affected by population history, a simple one-size-fits-all genotype-counting approach to inferring ROH in multiple populations is ill-advised. Here, we present a model-based method and accompanying software package for inferring ROH in microarray-derived SNP genotype that incorporates population-specific parameters and genotyping error rates as described in Pemberton *et al.* (2012). Version 1.1.0 added a method for reweighting LOD scores by gaps between SNPs, the option to report ROH lengths in cM, and the ability to provide per-genotype error rates. Version 2.0.0 reads VCF input directly (Section 5.1.1), analyses each population in a file separately rather than pooling them (Section 5.3), calls runs on the shared sex chromosome instead of treating it as an autosome (Section 5.4), and counts classified genotypes inside and outside the calls (Section 5.5). It also identifies the two modes of the LOD score density by their sign when it chooses the automatic cutoff, can choose that cutoff within strata of individuals of similar depth (Section 5.1.2), drops masked regions (Section 5.1.3), and reports the evidence available for each individual (Section 5.2). It is not backward compatible with files written by v1.1.6a, and the change log in Section 6 describes the differences.

If you use GARLIC please cite the following articles,

ZA Szpiech, A Blant, TJ Pemberton. (2017) GARLIC: Genomic Autozygosity Regions Likelihood-based Inference and Classification. Bioinformatics doi: 10.1093/bioinformatics/btx102.

TJ Pemberton, et al. (2012) Genomic patterns of homozygosity in worldwide human populations. American Journal of Human Genetics, 91, 275 − 292.

2 Obtaining `GARLIC`

Windows users no longer need `ann_figtree_version.dll` or `figtree.dll` alongside `garlic.exe`, because `GARLIC` no longer links FIGTree or ANN.

`GARLIC` pre-built binaries and source code are available at `https://github.com/szpiech/garlic`. To compile from source, change to the `src/` directory and type `make`, which detects the platform from `uname`, so no editing is needed. The only external dependency is the zlib library (`http://www.zlib.net/`), which the system provides on Linux and macOS. Threading uses `std::thread` from the C++11 standard library, for which the Makefile passes `-pthread` on Linux, and GNU GSL and FIGTree are no longer used. On Windows, `GARLIC` is built in an MSYS2/MinGW environment (`https://www.msys2.org/`) with zlib from the MinGW toolchain (`mingw-w64-x86_64-zlib` under MSYS2's MINGW64), and the MinGW GCC must be configured with the POSIX threading model, which is what provides `std::thread` on that platform.

3 Basic Usage

A run needs genotypes, an assembly, a genotyping error rate and a name for the output. The bundled example is human SNP-array data on hg18 coordinates:

```
garlic --tped example.tped.gz --tfam example.tfam --build hg18 --error 0.001 --winsize 30
--auto-winsize --out example
```

A priori we do not know how large to make the window, so this uses the built-in search (`--auto-winsize`). The search begins at the `--winsize` value and increases, so a starting size is required, and `--auto-winsize` alone is an error. All output is named starting with `example` (`--out example`). On this data the search settles on 50 SNPs, and the run writes:

- `example.log`, the run's parameters and progress;
- `example.params.json`, every flag's effective value together with the window size, LOD cutoff, size-class boundaries and random seed the run resolved, which can be replayed with `--load-params`;
- `example.freq.gz`, the allele frequencies used;
- `example.50SNPs.kde`, the LOD score density at the selected window size, which is part of the file name;
- `example.roh.bed`, the calls.

An `example.error` file appears beside these only if something is actually written to it. Adding `--froh` writes `example.froh.tsv`, one row per individual giving the autozygous total and the fraction of the genome it covers.

The same run from a VCF needs no `--tfam`, since a VCF names its samples, but it does need `--pop` if the samples are not all one population:

```
garlic --vcf cohort.vcf.gz --pop cohort.pop --build hg38 --error 0.001 --winsize 30 --auto-wir
--out cohort
```

Both readers accept gzipped input directly, and a VCF may also be BGZF. See Section 5.1.1 for what `garlic` reads from a VCF and Section 5.3 for what happens when a file holds more than one population. In that case each population is analysed separately, and the output names carry the population label.

Two settings that earlier versions did not require are now needed. A chromosome that looks sex-linked is not analysed until `--sex-system` says what to do with it (Section 5.4). An assembly must also be named, with `--build`, a `--centromere` file, or `--no-centromere` if the assembly has no gap to avoid.
If the automatic window size selection algorithm fails, you can output the KDEs of the LOD score distribution for multiple window sizes (without calling ROH) by using the `--winsize-multi` argument, i.e.

```
garlic --tped example.tped.gz --tfam example.tfam --build hg18 --error 0.001 --winsize-multi
30 40 50 60 70 80 90 --out example
```

This will generate KDEs for your inspection. Once you've chosen a window size, you should rerun garlic specifying that size. For example, if you choose a window size of 60 SNPs, then you would run

```
garlic --tped example.tped.gz --tfam example.tfam --build hg18 --error 0.001 --winsize 60
--out example
```

The automatic cutoff is the minimum of the window LOD score density, pooled over the cohort, between its two modes. The non-autozygous mode is the tallest peak below zero and the autozygous mode is the tallest peak above zero (`manual/lod-mode-rule.tex` gives the derivation). A cohort without long runs still has a small mode above zero from short runs of background autozygosity, and gets a cutoff that calls only short runs (a simulated cohort whose chromosomes were built from independent 1-Mb blocks at an effective size of 200 000 was called at 1.2% of the genome, in runs of at most 261 kb). A density with no

peak above zero therefore means that the windows carry too little evidence, not that the cohort carries no autozygosity. Examples are genotypes called from one or two reads under a population prior and a cohort of mixed depth pooled without `--stratify-cutoff`. Such a run completes with no calls and a warning, and `.params.json` records the verdict with `lod_cutoff` null. A density with no peak below zero stops and asks for `--lod-cutoff`. A cohort of a handful of individuals is still a case for a cutoff carried over from a comparable, larger cohort rather than for the automatic rule.
If you already know what LOD score cutoff to use (say you are analyzing more individuals from a previously studied population), you can use the `--lod-cutoff` argument, i.e. if your LOD score cutoff is known to be 2.5 then you would run

```
garlic --tped example.tped.gz --tfam example.tfam --build hg18 --error 0.001 --winsize 60
--out example --lod-cutoff 2.5
```

If you already know what size thresholds to use for size classification (say you are analyzing more individuals from a previously studied population), you can use the `--size-bounds` argument, i.e. if your size thresholds are known to be 500000 and 1000000 for the boundaries between short/med and med/long, respectively, then you would run

```
garlic --tped example.tped.gz --tfam example.tfam --build hg18 --error 0.001 --winsize 60
--out example --lod-cutoff 2.5 --size-bounds 500000 1000000
```

Supplying both is also how to reproduce an analysis made with an earlier version, whose automatic cutoff selection differed. The values to pass are in that run's log.

To run the wLOD, you must provide a map file and give the `--weighted` flag:

```
garlic --tped example.tped.gz --tfam example.tfam --map example.map.gz --weighted --build
hg18 --winsize 60 --out example --error 0.001
```

To count classified genotypes (deleterious against tolerated, for example) inside and outside the calls, add `--features` with a file classifying the sites (Section 5.5):

```
garlic --tped example.tped.gz --tfam example.tfam --build hg18 --error 0.001 --winsize 60
--lod-cutoff 2.5 --size-bounds 500000 1000000 --features classes.txt --out example
```

Other command line arguments are described below, and all of them are listed with their types and defaults in Appendix A.

4 Statistics implemented

We previously advocated the application of a logarithm of the odds (LOD) score measure of autozygosity applied in a sliding-window framework to infer ROH in high-density SNP genotype data (Pemberton *et al.*, 2012). Fundamentally, for SNP k in individual i, this method calculates the log-likelihood ratio of observed genotype $G_{i,k}$ under the hypotheses of autozygosity and non-autozygosity, incorporating an assumed genotype error rate ϵ and population-specific allele frequencies. The LOD score of window w in

individual i is then calculated as the sum of the log-likelihood ratios of the K SNPs in the window:

$$LOD(w,i) = \sum_{k=1}^{K} \log_{10}\left(\frac{Pr[G_{i,k}|X_k = 1]}{Pr[G_{i,k}|X_k = 0]}\right). \quad (1)$$

Here, $Pr[G_{i,k}|X_k = 1]$ is the probability of observing genotype $G_{i,k}$ under the hypothesis of autozygosity ($X_k = 1$), and $Pr[G_{i,k}|X_k = 0]$ is the probability of observing genotype $G_{i,k}$ under the hypothesis of non-autozygosity ($X_k = 0$). For a biallelic locus with alleles A and B that have population frequencies p_A and p_B and a genotype error rate ϵ, the genotype probabilities under the autozygosity and non-autozygosity hypotheses are given in Table 1.

Calculating $LOD(w,i)$ for all windows in all individuals in a given sample set, examination of the distribution of scores shows clear bimodality, with windows in the left-hand mode supporting the hypothesis of non-autozygosity and those in the right-hand mode supporting the hypothesis of autozygosity. The area under the autozygous mode decreases with increasing window size until it disappears, potentially reflecting the window size beyond which window length is often longer than ROH length leading to the inclusion of non-autozygous regions in the $LOD(w,i)$ calculation that mask the presence of autozygosity. A logical window size to use for ROH detection is therefore the largest window size where the distribution of LOD(w,i) is bimodal, with windows defined as autozygous if their $LOD(w,i)$ is greater than the local minimum between the two modes. Contiguous autozygous windows are subsequently joined to define ROH.

If the weighted flag is used, wLOD scores will be calculated as

$$wLOD(w,i) = \sum_{k=1}^{K} \log_{10}\left(\frac{Pr[G_{i,k}|X_k = 1]}{Pr[G_{i,k}|X_k = 0]}\right) \times Corr(p_k, [p_1, p_K]) \quad (2)$$

$$\times\, Pr[norecombination|[g_{k-1}, g_k]] \quad (3)$$

$$\times\, Pr[nomutation|\mu, [p_{k-1}, p_k]], \quad (4)$$

where

$$Corr(p_k, [p_1, p_K]) = \frac{1}{\sum_1^K LD_{k,i}}, \quad (5)$$

$$Pr[norecombination|[g_{k-1}, g_k]] = e^{-2M(g_k - g_{k-1})}, \quad (6)$$

and

$$Pr[nomutation|\mu, [p_{k-1}, p_k]] = e^{-2M\mu(p_k - p_{k-1})}. \quad (7)$$

$LD_{k,i}$ is computed as r^2 if data are phased and as HR^2 if data are unphased. g_i and p_i are the genetic distance and physical distance at marker i, respectively.

Table 1: Probability model for genotypes under autozygosity and non-autozygosity.

Observed genotype G_k	$Pr[G_{j,k}\|X_k = 1]$	$Pr[G_{j,k}\|X_k = 0]$
AA	$(1-\epsilon)p_A + \epsilon p_A^2$	p_A^2
AB	$2\epsilon p_A p_B$	$2p_A p_B$
BB	$(1-\epsilon)p_B + \epsilon p_B^2$	p_B^2
Missing	1	1

5 Program Options

Using the command line flag `--help`, will print a help dialog with a summary of each command line option. The full reference is reproduced in Appendix A, which `make docs` generates from the same help strings.

5.1 Input Files

Genotypes are read from a VCF (`--vcf`), a TPED/TFAM pair (`--tped`/`--tfam`, see `http://pngu.mgh.harvard.edu/~purcell/plink/data.shtml#tr`). The two are mutually exclusive. Either may be read gzipped (`http://www.gzip.org/`) without decompressing first, and a VCF may also be BGZF, as written by `bgzip` or `bcftools`. Loci are assumed to be in order of physical position, and all records for a chromosome must be contiguous, as `bcftools sort` produces. If the file contains more than one population, each is analysed separately (Section 5.3). `--pop` overrides the labels, which for a TPED/TFAM pair are otherwise taken from column 1 of the TFAM.

5.1.1 VCF input

Biallelic SNVs carrying a `GT` field are used. Multiallelic sites, indels, symbolic ALT alleles (`*`, `<NON_REF>`) and records with no `GT` in `FORMAT` are skipped, and the number of each is written to the log. The `FILTER` column is reported but acted on only under `--vcf-pass-only`, since it records the variant caller's thresholds rather than `garlic`'s.

A VCF states things that a TPED cannot, and `garlic` checks them. Ploidy is given per call, so anything other than a diploid genotype is an error that names the site and the sample, except on a declared sex chromosome (Section 5.4). Phase is also given per call, so `--phased` requires `|` in every call. Samples are named in the `#CHROM` line, so `--tfam` is not accepted with `--vcf`. Use `--pop` for population labels. Without it every sample is labelled `unknown` and `garlic` warns, because the allele frequencies are then pooled over all samples. With `--pop`, each population is analysed separately (Section 5.3).

The counted allele is ALT. For the same genotypes a TPED counts whichever allele it saw first, so the frequency column may come out as $1 - f$ relative to a TPED run. This changes no ROH call, and a `.freq.gz` file can be used with either reader, because `garlic` compares the file's `ALLELE` column against its own coding and flips the frequency when they differ.

Genetic map positions are not part of a VCF, so `--weighted` and `--cm` still require `--map`.

Per-genotype qualities are read from the `FORMAT` column with `--gl-type`. This is the only correct way to use a VCF's `PL` or `GL`. A VCF normalises both so that the called genotype's value is exactly 0 for every call, confident or not, and the information is in how much worse the alternatives are. `garlic` therefore reads the whole array and computes

$$P(g) \propto 10^{-\mathrm{PL}_g/10}, \qquad \text{error} = 1 - \frac{P(\text{called})}{\sum_g P(g)},$$

with `GL` entering the same expression as $\mathrm{PL} = -10\,\mathrm{GL}$. This reproduces the `GQ` that encodes the same confidence (a `PL` of $(0, 30, 60)$ gives 0.001, exactly `GQ` 30). If the requested field is absent the run stops, and the message names whichever of `GQ`, `PL` and `GL` the record does carry.

Low-coverage sequence data. None of the per-genotype quality forms above is suitable for low-coverage sequencing (below roughly 4×). Each of them gives one error rate per genotype and is evaluated at the called genotype, so an uncertain homozygous call still counts as evidence for autozygosity, only with a smaller weight. At low depth the dominant error is allelic dropout, in which the reads sample only one of a heterozygote's two alleles. Dropout turns heterozygotes into homozygotes, and a run of such calls is what autozygosity looks like to the LOD score. In a simulation of independent sites with reads drawn at 0.5–1× (60-SNP windows, 10% of the genome autozygous), a per-genotype error taken from `GQ` called essentially the whole genome autozygous. In the same simulation, hard calls made under a Hardy-Weinberg prior with a run-wide `--error` degraded more gracefully but lost the automatic cutoff at 2×.

The companion script `vcf2tped.pl` is deprecated. It writes a TPED and a `tgls` file, and a `tgls` file holds one value per genotype, which cannot represent a normalised `PL` or `GL`. `garlic` rejects such a file.

5.1.2 Cohorts of mixed depth: `--stratify-cutoff`

The automatic LOD score cutoff is the valley between the two modes of the window LOD density pooled over every individual. This is appropriate when every individual carries the same evidence per window. However, an individual at low depth, or with many missing genotypes, has both of its modes compressed toward zero, so a cutoff placed for the cohort calls most of its genome autozygous.

The location of an individual's non-autozygous LOD mode is a precise measure of the evidence its windows carry. On the bundled cohort it varies by 1.3 LOD between individuals. `--stratify-cutoff` sorts the individuals by that mode, joins neighbours whose modes differ by less than a quarter of the smaller spread into a stratum, and applies the pooled rule within each stratum. A cohort of uniform evidence is one stratum and gets exactly the pooled cutoff (the test suite checks that the calls are byte-identical). A mixed cohort splits by depth, and each stratum's valley rests on tens of thousands of windows rather than on one individual's.

The per-individual mode, spread, stratum and applied cutoff are written to `<out>.ind.tsv`. The `.kde` file is the largest stratum's density, and `.params.json` records `lod_cutoff` as the number of strata and flags it as such. A stratum of fewer than three individuals does not get a valley of its own and takes the largest stratum's cutoff, with a warning. One or two individuals' windows are too few for the rule, and such a stratum is as likely to be an individual whose lowest substantial mode is its autozygous one (more than half autozygous on the analysed chromosomes, as is one of the 45 individuals on chromosome 21 of the bundled data). A stratum whose density has no valley borrows the median standardised valley of the other strata, through its own mode and spread. The flag needs the automatic cutoff and does not subsample.

Without the flag, `garlic` still locates the modes and warns when the cohort would form more than one stratum. The warning names the strata, so a mixed cohort analysed with one cutoff is a visible choice.

5.1.3 Reference and mapping bias: `--exclude-regions`

Reads carrying the non-reference allele map less often and with lower quality, so at a true heterozygote the fraction of ALT reads is below one half. In this model that is a *directional* error (a heterozygote observed as a reference homozygote), and a run of such calls is what autozygosity looks like. The bias has two parts, and they are handled differently.

Regional bias (repeats, segmental duplications, structurally divergent segments) mis-maps reads wholesale, and no per-site model recovers from it. `--exclude-regions <bed>` drops every site inside the regions of a BED file before anything is computed. The file may be a mappability mask, a repeat annotation, segmental duplications, or any set of regions in which calls are not to be trusted. It is standard BED (0-based half-open, further columns ignored, `track`, `browser` and `#` lines skipped), gzipped or not. The mask applies on every chromosome, autosomes included. Chromosomes in the file that are not in the data are ignored, since a genome-wide mask names them all. A file that covers no site at all draws a warning, since that usually means the wrong assembly or chromosome naming. The dropped sites are removed from the frequency file and from every window. The F_{ROH} denominator is not reduced by them, unlike with `--par`, because a run of homozygosity can span a short masked region. `--freq-only` does not apply the mask and refuses the combination.

Read-level bias , an ALT read fraction below one half at heterozygotes everywhere, is not corrected by `garlic`. It is best reduced before the data reach `garlic`.

- Map to a reference that carries the alternative alleles (an alt-aware or pangenome mapper, or a population-matched reference).

- Use ANGSD's `-baq 1 -C 50` to down-weight mis-aligned bases, and its `-minMapQ` and `-minQ` filters.
- Filter short reads, whose mapping is the most reference-biased.
- For ancient DNA, rescale damaged bases (`mapDamage`) or drop transitions (`-rmTrans 1`), which also removes the deamination signal that mimics heterozygosity.
- Take allele frequencies from a source that is not itself biased (ANGSD's `-doMaf`, or `--freq-file` from the high-coverage members of the cohort).

5.1.4 File formats

tped/tfam (required):

`http://zzz.bwh.harvard.edu/plink/data.shtml#tr`

map (required when `--weighted` or `--cm` are set):

```
<chr> <snpid> <genetic pos> <physical pos>
http://zzz.bwh.harvard.edu/plink/data.shtml#map
(negative positions will cause errors)
```

tgls (optional and legacy, see the note on `vcf2tped.pl` above):

```
<chr> <snpid> <unused> <physical pos> <GL ind1> <GL ind2> ... <GL indN>
```

One value per genotype. Under `--gl-type PL` or `GL` these are read as the probability that the genotype is *correct*, which is not what a VCF's `PL` and `GL` fields hold, and a file of VCF-derived values is rejected with an error. `--gl-type GQ` is the one form whose VCF values can be used here directly.
exclude-regions (optional; see §5.1.3):

```
<chrom> <chromStart> <chromEnd> [ignored ...]
```

Standard BED, 0-based half-open. `track`, `browser` and `#` lines are skipped.
pop (optional, and with `--vcf` the only way to set population labels):

```
<sample id> <population> [sex]
```

Matched by sample ID, never by position. Note that the column order is the opposite of a TFAM's first two columns, and a TFAM-ordered file is diagnosed as such. The optional third column uses PLINK sex coding. On a declared sex chromosome a recorded sex is checked against the heterozygosity `garlic` measures there, and the run stops if more than half the individuals contradict it (Section 5.4).
freq (optional; written by every run, or on its own with `--freq-only`):

```
CHR SNP POS ALLELE FREQ
<chr> <locus ID> <pos> <allele> <freq>
```

With more than one population there is one frequency column per population, headed by its label, in place of the single `FREQ` column. A file carrying just one such column is applied to every population, whatever that column is called (Section 5.3).
centromere file (one of `--build`, this file, or `--no-centromere` is required):

```
<chr> <centromere start> <centromere end>
```

5.1.5 --tped

Use `--tped <string>` to specify a .tped (transposed PLINK; Purcell *et al.* (2007)) file (see `http://zzz.bwh.harvard.edu/plink/data.shtml#tr` for exact specifications) containing genetic variant information.

5.1.6 --tped-missing

Use `--tped-missing <char>` to specify the missing data code in the .tped file. Default is `0`.

5.1.7 --tfam

Use `--tfam <string>` to specify a .tfam (transposed PLINK; Purcell *et al.* (2007)) file (see `http://zzz.bwh.harvard.edu/plink/data.shtml#tr` for exact specifications) containing sample information.

5.1.8 --tgls

Use `--tgls <string>` A tgls file containing per-genotype likelihoods. Formatted: <chr> <snpid> <unused> <physical pos> <GL ind1> <GL ind2> ... <GL indN>. Default: none.

Per-genotype likelihoods should be given as either a phred-scaled (`--gl-type PL`) or a $\log_{10}$-scaled (`--gl-type GL`) representing the probability that the genotype is correct or a phred-scaled likelihood that the genotype is wrong (`--gl-type GQ`). For example, if $p = 0.999$ is the probability that the genotype is correct, then $PL = -10\log_{10}(p) = 0.00434511774018$, $GL = \log_{10}(p) = -0.000434511774018$, and $GQ = -10\log_{10}(1-p) = 30$.

5.1.9 --centromere

Use `--centromere <string>` to specify a file containing custom centromere boundaries formatted as

```
<chr> <start> <end>
```

5.1.10 --freq-file

`--freq-file <string>` A file specifying allele frequencies for all variants. File format:

```
CHR      SNP      POS      ALLELE   FREQ
<chr>    <locus ID>        <pos>    <allele>         <freq>
```

By default, this is calculated automatically from the provided data.

5.1.11 --map

`--map <string>` gives a scaffold genetic map. Sites that are not in the file are interpolated, and sites outside its bounds are filtered. The map is required for wLOD calculations and for reporting ROH in units of cM. Its format is given under File formats above, and negative positions cause errors.

5.2 Output Files

`garlic` produces several files as output. The `.log` file records the runtime parameters, and `.error` records any errors that occur. `.error` is created only if something is written to it. A `.params.json` file records the effective value of every flag together with the values resolved during the run (selected window size, LOD cutoff, size class boundaries and random seed), and it can be replayed with `--load-params`.

A gzipped `.freq` file will be output giving the allele frequencies for each locus in the dataset. Formatted as

```
CHR     SNP     POS     ALLELE  FREQ
<chr>   <locus ID>      <pos>   <allele>        <freq>
```

With more than one population the file carries one frequency column per population, headed by its label:

```
CHR     SNP     POS     ALLELE  POPA    POPB
<chr>   <locus ID>      <pos>   <allele>        <freq>  <freq>
```

This is the format `--freq-file` reads, so a run's own frequency file can be given back to `garlic` to reproduce that run. A file with a single frequency column is applied to every population, whatever the column is called, so a frequency file from an earlier version can still be used. `--freq-only` writes the same file without performing the analysis.

With `--froh`, a `.froh.tsv` table gives each individual's autozygous total and the fraction of the genome it covers, one row per individual with the size classes across the columns. The denominator is stated in the header, as are the size-class boundaries actually used, so the fractions can be recomputed or compared across runs.

`--froh-denominator` chooses what that fraction is taken of. `analyzed`, the default and what `garlic` has always done, is the span the markers cover, from the first to the last locus of each analysed chromosome, less the centromere and any region excluded with `--par`. This is what a marker set can support on its own, but it moves with the marker set, so F_{ROH} values from two different arrays are not directly comparable. `assembly` is the full length of each analysed chromosome, with nothing subtracted (not the centromere, the pseudoautosomal regions or the part beyond the outermost marker). Published F_{ROH} denominators are of this kind. For example, the hg19 autosomes sum to 2,881,033,286 bp and chrX is 155,270,560 bp, which are the 2881.03 Mb and 155.27 Mb of Cotter *et al.* (2024). On the bundled example data the two denominators differ by about 5%.

The sum is over the chromosomes that were *analysed*, and the assembly supplies each one's length. A run restricted with `--chr` is divided by those chromosomes alone, and a contig that carries no data is never looked up, so a VCF full of decoys, unplaced scaffolds and alternate haplotypes needs no filtering here. A chromosome that was analysed and has no length available stops the run rather than falling back to its marker span, since mixing the two conventions across chromosomes gives a denominator that is neither.

Lengths come from `--chr-lengths`, from `--build`, or from a VCF's `##contig` header, in that order of precedence. `--chr-lengths` takes two whitespace-separated columns, chromosome and length, and also accepts a `samtools faidx .fai` unchanged, since its first two columns are already name and length. It is the way to use this mode on a non-human assembly and on TPED input, where nothing in the data states a length. The built-in tables are the GRC assembly reports' own numbers, recorded with their source in `centromeres/chr_lengths.txt`. `assembly` is refused with `--cm`, because a chromosome's length in base pairs is not its length in centimorgans.

Whenever lengths are available, `garlic` checks that no marker lies beyond the end of its chromosome. This is how a wrong `--build` shows itself, and nothing else in the program can detect it. The check matters in either mode, because the centromere and pseudoautosomal coordinates in use came from the

same assembly. Under `assembly` a violation stops the run, since those lengths are the denominator. Under `analyzed` they feed nothing, so it is a warning. Where a VCF header and `--build` both state a length and disagree, one of them is not this assembly, and `garlic` reports the conflict rather than choosing one.

Kernel density estimation of the LOD score distribution is output to `.<winsize>SNPs.kde` formatted as

```
<LOD score> <density>
```

ROH calls are output to `.roh.bed`, which are formatted as UCSC BED files. There is one track per individual (all individuals in the same file) and the data are formatted as

```
<chr> <ROH start> <ROH end> <size class> <ROH length> <placeholder> <
   placeholder> <placeholder> <RGB track color>
```

Following the BED specification, `<ROH start>` is 0-based and `<ROH end>` is exclusive, so `<ROH end>` minus `<ROH start>` equals `<ROH length>`. Note that under `--cm` the boundaries remain physical positions while `<ROH length>` is reported in cM, so the identity does not hold in that mode. Versions before 2.0.0 wrote `<ROH start>` as a 1-based position, so coordinates from those versions are shifted by one relative to these. When requested, raw LOD scores are also output. This file contains one row per individual (corresponding to the order in the `.tfam` file), and LOD scores for each window are given on the columns.

```
<LOD score window 1> <LOD score window 2> ... <LOD score window N>
```

The per-individual information report. Every run also writes `<out>.ind.tsv` (per population when there are several), with one row per individual. A window's LOD is a sum over the sites in it, and a site that carries no information for an individual (a missing genotype) contributes exactly 0. Two individuals with the same window size can therefore be summing very different amounts of evidence. The report records, per individual:

- `n_sites`, the analysed sites on chromosomes where the individual is diploid;
- `frac_informative`, the fraction of them with a called genotype;
- `snps_per_window_median` and `_p05`, the median and 5th percentile of informative sites per window;
- `n_het_expected`, the sum of $2p(1-p)$ over the sites used for `het_obs_exp`;
- `het_obs_exp`, observed over expected heterozygotes from called genotypes.

The last two are computed over the sites in windows whose LOD is below 0, which stand in for non-autozygous windows without needing the cutoff, so they are not confounded by the individual's own autozygosity. `garlic` warns when an individual's 5th percentile falls below half the window size.

`het_obs_exp` is the ordinary sample-quality statistic for called data, and it is not a low-depth estimator. At low depth it reads well below 1 even without any bias, because the maximum-likelihood call at one or two reads is a homozygote. Its sampling noise is roughly $1/\sqrt{\texttt{n_het_expected}}$, a few hundredths on one chromosome and a few thousandths genome-wide, so it should be read against the cohort rather than as an absolute. It corrects nothing, and the report is there so that an individual whose data differ from the rest of the cohort can be seen.

5.2.1 `--out`

Use `--out <string>` to provide a base name for an output file. This will be used in place of `<outfile>` above. Default value is `outfile`.

5.3 Multiple populations

The LOD score is parameterised by population allele frequency, so pooling distinct populations inflates heterozygosity relative to any one of them and biases the ratio for all of them. When the input contains more than one population label, `garlic` therefore analyses each population separately and in turn. Each receives

- allele frequencies computed from its own individuals;
- its own site filtering, so a site that is monomorphic in one population but polymorphic in another is dropped where it is uninformative;
- its own automatically selected window size, LOD score cutoff and size-class boundaries;
- its own random number stream, derived from `--seed`, so its result does not depend on how many populations preceded it in the file.

A value given explicitly on the command line is not re-derived, so `--winsize`, `--overlap-frac`, `--lod-cutoff` and `--size-bounds` apply to every population.

Analysing several populations together therefore gives each of them exactly what analysing it alone would give. The populations do not interact, so running them in one command is only a convenience.

Populations are processed one at a time, so peak memory is bounded by the largest population rather than by their sum.

5.3.1 Where the labels come from

For a TPED/TFAM pair the label is column 1 of the TFAM. PLINK calls that column the family ID, and in many files it holds one family, or even one sample, per value. Such a file produces one analysis per family, which is almost certainly not what was intended. `garlic` warns when the smallest population has fewer than five individuals, and again when there are fewer than two individuals per population on average, but it cannot tell the two situations apart with certainty. Use `--pop` to supply real population labels, or `--pool-populations` to ignore them.

A VCF carries no population labels, so `--pop` is the only way to set them; without it every sample is labelled `unknown` and the analysis is pooled.

5.3.2 Output files

With one population the output file names are unchanged. With more than one, the population label is inserted before the extension: `<out>.<POP>.roh.bed`, `<out>.<POP>.froh.tsv`, `<out>.<POP>.ind.tsv`, `<out>.<POP>.params.json` and `<out>.<POP>.<winsize>SNPs.kde`. The `.log`, `.error` and `.freq.gz` files are written once for the whole run.

5.3.3 `--pool-populations`

This flag makes `garlic` ignore the labels and analyse every individual together, as it did before per-population analysis. Allele frequencies, the LOD score cutoff, the window size and the size classes are then computed over every sample at once, and the unlabelled output names are restored. Use it when the labels are not populations, or to reproduce a run from an earlier version. The `.froh.tsv` header and `.params.json` both record that the populations were pooled.

5.3.4 When a population cannot be analysed

A small population can fail size-class fitting or cutoff selection where a pooled fit would have succeeded. `garlic` reports which population failed and stops, rather than writing a partial set of results that looks complete. Supplying `--size-bounds` or `--lod-cutoff` avoids the automatic step that failed.

5.4 Sex chromosomes

A hemizygous genotype is written as a homozygous call, so a sex chromosome analysed as an autosome is called as one run spanning its whole length in every heterogametic individual, and any F_{ROH} computed from that is inflated. `garlic` therefore refuses to analyse a chromosome that looks sex-linked until it is told what to do with it.

5.4.1 Declaring the system

`--sex-system` takes `xy`, `zw` or `none`. It supplies the one thing that genotype data cannot, which is the sex *code* that is heterogametic. PLINK codes 1 male and 2 female whatever the species' sex determination system is, so in a ZW species the *homogametic* sex is the one coded 1. Leaving the flag unset is not the same as `none`, because unset means nothing has been said and the run stops, while `none` asserts that every chromosome is diploid in every individual.

The conventional names follow from the system (X and Y under `xy`, Z and W under `zw`, plus M or MT for the mitochondrion). A chromosome with any other name is declared explicitly with `--sex-chr` (diploid in the homogametic sex), `--sex-chr-degenerate` (carried only by the heterogametic sex) or `--haploid-chr`. A chromosome's role does not depend on the system, so X and Z are both the shared chromosome and Y and W are both the degenerate one. Only the shared chromosome is analysed. The others cannot carry a run of homozygosity in anybody, and they are dropped with a log line.

The numbers 23, 24, 25 and 26 are X, Y, the pseudoautosomal region and the mitochondrion under PLINK's coding of *human* data, and ordinary autosomes in any species with that many chromosomes. `garlic` resolves them only when `--build` names a human assembly, since that already asserts human coordinates. Without one it refuses rather than picking a reading.

5.4.2 What the analysis does

Allele frequencies on the shared sex chromosome are computed over *everyone*, with the heterogametic sex contributing one allele and the homogametic sex two. Runs of homozygosity are then called only in the individuals who are diploid there. The LOD score cutoff, the window size and the size-class boundaries are estimated on the autosomes and applied to the sex chromosome, and `garlic` reports what the sex chromosome's own windows would have given. This follows Cotter *et al.* (2024), who give the reason for the cutoff: one chromosome's windows from part of a cohort usually cannot locate a second minimum in the density.

This is the default rather than the only option. When the reported diagnostic shows that the autosomal cutoff is badly placed on the sex chromosome, another cutoff can be given with `--sexchr-lod-cutoff`,

and a different window size with `--sexchr-winsize`. Both apply to the shared sex chromosome alone and leave every autosomal call where it was. Because the estimate on that chromosome usually fails, these flags take values rather than asking `garlic` to re-estimate, and the diagnostic is where the values come from. A LOD score cutoff is specific to the window size it was estimated at, so `--sexchr-winsize` requires `--sexchr-lod-cutoff`. The values that were applied, whether or not the flags were given, are recorded in `<out>.params.json` as `sexchr_winsize` and `sexchr_lod_cutoff`.

A half call carries no genotype. The LOD score takes its default branch for one, so `--tgls` and a VCF's `GQ`, `PL` or `GL` need no special handling, since the per-genotype error rate is never read there. This matters for a hemizygous VCF, where a haploid `PL` array holds one value per allele (two at a biallelic site rather than three), because the array is never parsed. Both LD statistics behind `--weighted` also skip an uncalled genotype, so the LD matrix on the shared sex chromosome is estimated among the individuals who are diploid there. Under `--ld-subsample` the subsample for that chromosome is drawn from those individuals, so the number requested is the number that informs it.

Who is diploid there is established by a check that always runs. Each individual's heterozygosity on that chromosome is compared against `--het-rate-bounds` (0.02 and 0.10 by default). Below the first bound the individual reads as hemizygous, above the second as diploid, and between them as ambiguous, in which case it is neither called there nor counted towards the allele frequency. A recorded sex is checked against this, and an unrecorded one is inferred from it. The per-individual result is written to `<out>.sexcheck.tsv`. Heterozygous calls in an individual with one copy of the chromosome are impossible, so they are counted and discarded. If more than half the individuals with a recorded sex contradict it, `garlic` stops, because either `--sex-system` is inverted or the chromosome was called as diploid, and continuing would discard every one of those calls.

F_{ROH} is reported separately for the autosomes and for the sex chromosome, each with its own denominator, and is `NA` rather than zero for an individual that cannot be autozygous there. The two are not summed. The X coalesces faster than the autosomes and carries more homozygosity at the same level of consanguinity, so the two quantities should be compared rather than pooled.

5.4.3 Pseudoautosomal regions

A pseudoautosomal region is diploid in both sexes. `--par` takes one or more `<chr>:<start>-<end>` intervals, and `--par-file` reads them from a file. Both accept any number of regions per chromosome (humans have two), and the two flags accumulate. Unlike `--centromere`, repeated rows for one chromosome add a region rather than replacing one. The loci inside are dropped, each region's count is reported separately, and a region that matches no locus draws a warning that names it.

They are identified by coordinates on a declared sex chromosome and never by a chromosome code, because PLINK's 25 is an ordinary autosome in most species.

For the human X, `--build` supplies the regions. The coordinates are the Genome Reference Consortium's own definition for that assembly, recorded with their source in `centromeres/par_regions.txt`, and they are filed under whatever name the data gives that chromosome (`chrX` in one file and `23` in the next). Naming any region with `--par` replaces them, and `--par none` turns them off. No table is supplied for any other shared sex chromosome.

Running without the regions, on a non-human assembly or under `--par none`, has a bounded cost. The heterogametic sex's genotypes in a pseudoautosomal region are recoded as hemizygous along with the rest of the chromosome. Their real heterozygous calls there are therefore counted as impossible and discarded, which inflates `n_het_recoded` by roughly that region's share of the markers (about 1–2% of the human X). Their contribution to the allele frequency in the region is also biased, because dropping heterozygotes while keeping homozygotes is not a random thinning. Both effects are bounded by the total pseudoautosomal share of the chromosome, which is larger in a species that has several such regions.

5.4.4 Chromosome names

Names are matched case-insensitively and ignoring a leading `chr`, so `chrX`, `chrx`, `CHRX`, `X` and `x` are one chromosome, and so are `1`, `01` and `chr1`. Output keeps whatever the input said. Two chromosomes in one file whose names differ only in this way are an error. So is the same name appearing twice, which means that one chromosome's rows are not contiguous. `garlic` starts a new chromosome whenever the chromosome column changes and would otherwise analyse it as two.

5.5 Counting classified genotypes

A common use of a set of runs of homozygosity is to ask whether genotypes of some class (deleterious, non-synonymous, or whatever an annotation supplies) are found inside them more often than outside. `--features` does that counting as part of the run that makes the calls, and writes `<out>.counts.tsv`.

The classification is a file of four whitespace-separated columns, the chromosome, the position, the allele that carries the classification, and the class label:

```
#chr    pos         allele  class
chr21   15481365    T       probably_damaging
chr21   15481365    T       nonsynonymous
21      15502312    A       benign
```

The label is arbitrary text, and there may be any number of distinct labels. A site may appear more than once, as the first two rows above do, so two annotation schemes can classify the same variant, and the categories need not be mutually exclusive. The same class twice at one site is an error. Chromosome names are matched ignoring case and a `chr` prefix, and the allele is compared as a string, so an indel allele from a sequenced call set works as well as a single base.

For every individual and every class, the table gives three numbers in each bucket. `hom` is the number of genotypes homozygous for the classified allele, `het` the number carrying one copy, and `n` the number of classified sites at which that individual has a genotype at all. The last is the denominator, and it separates "more deleterious homozygotes inside runs" from "more sites inside runs". The buckets are the size classes the population was analysed with, then `NONE`, then `UNASSESSED`, then `ALL`. The count inside runs of homozygosity is `ALL` less `NONE` less `UNASSESSED`.

`UNASSESSED` counts sites where no run could have been called for anybody: a chromosome that was not analysed, one carrying fewer than two loci, an assembly gap, a region excluded by `--par`, or a position outside the span the markers cover. It is kept apart from `NONE` because folding the two together inflates the outside-runs count by however much of the genome the analysis left out. Two further columns account for the rest of each individual's classified sites. `n_hemi` counts sites on a chromosome where that individual is not diploid, and `n_missing` counts sites it has no call for. Every classified row lands in exactly one of these places, so $\sum \texttt{ALL_n} + \texttt{n_missing} + \texttt{n_hemi}$ is the same for every individual the genotypes covered.

The genotypes are read from the file without filtering. The filtered matrix that `garlic` calls from has had monomorphic sites dropped population by population, has been pruned by `--chr` and `--autosomes-only`, and has had excluded regions removed, and counting from it would discard exactly the rare functional sites the question is about. By default the input the run was called from is read a second time. When the classified variants are in another file (calls from a genotyping array and classifications from sequencing, for example), give that file with `--tped-counting` and `--tfam-counting`, or with `--vcf-counting`. The two files need not share a single site. Individuals are matched by identifier, and one present in only one of the two files is reported rather than counted as having no runs.

5.5.1 Counting against calls that already exist

`--roh-file` counts against a `.roh.bed` written earlier, or published, instead of calling anything. It needs `--features` and one of the counting genotype flags, and it accepts no calling input.

Less can be known this way, and the table records it. A `.roh.bed` records where the runs are, but not which chromosomes were analysed, how far the markers reached, or what the size class letters mean. So nothing can be marked `UNASSESSED`, every site outside a run is `NONE`, and the size class boundaries are reported as unknown. Counting inside the run that makes the calls knows all three, so it is preferable where there is a choice.

Note that version 1.1.6a was released with two different `chromStart` conventions under the same version string, so the version in a file's track line does not say which one it uses. `garlic` works it out from the file instead. A conforming BED has `chromEnd` − `chromStart` equal to the length column, and the older convention has it one less. The convention is determined from every call rather than from the first, a file that is half each is refused as two files concatenated, and the convention used is recorded in the output header. Under `--cm` the length column is a genetic distance and settles nothing, so that case draws a warning that names its assumption.

5.6 Controlling GARLIC

The remaining flags control window construction, LOD score calculation, cutoff selection, size classification, reproducibility and output. They are listed with their types and defaults in Appendix A, which `make docs` generates from `garlic`'s own help strings.

6 Change Log

```
2.0.0 - Correctness, performance, and command line work.

--features counts classified genotypes inside and outside the calls,
   replacing the
count_features_in_roh.pl and count_damaging_genotypes_in_roh.pl scripts,
   which are removed.
See "Counting classified genotypes" under Program Options. Any number of
   arbitrary class
labels, a site may carry several, and the table reports hom, het and a
   denominator n per
(class, size class) with a separate UNASSESSED bucket for sites where no
   run could have been
called. --roh-file counts against a .roh.bed that already exists.

A sex chromosome is no longer analysed as an autosome. A run containing one
    stops until
--sex-system says which sex code is heterogametic. See "Sex chromosomes"
   under Program Options.
Frequencies there come from everyone with the heterogametic sex
   contributing one allele, calling
is restricted to the individuals who are diploid, the cutoff and size
   classes are estimated on
```

the autosomes, and FROH is reported for the two regions separately with NA where an individual
cannot be autozygous. The FROH table is now one row per individual. --autosomes-only drops every
non-autosomal role rather than four names, and 23-26 are resolved only under a human --build.
--par excludes pseudoautosomal regions by coordinate, any number of them. A hemizygous VCF is
readable on a declared sex chromosome. Chromosome names match case-insensitively and ignoring a
chr prefix.

Input with more than one population label is now analysed one population at a time (see
"Multiple populations" under Program Options). Each population gets allele frequencies from its
own site filtering, its own automatically selected window size, LOD cutoff and size-class
boundaries, and its own random stream. A value given on the command line is not re-derived
and applies to every population. Analysing several populations together gives each of them
what analysing it alone gives.

With ONE population nothing changes. With more than one, the label is inserted before the
extension: <out>.<POP>.roh.bed and so on. --pool-populations restores the previous behaviour
exactly, and its output is byte-identical to the previous version's.

Column 1 of a TFAM is garlic's population label but PLINK's FAMILY ID, so a file that uses it
per family will now produce one analysis per family. garlic warns, but use --pop or
--pool-populations.

--freq-file accepts one frequency column per population, matched by header, and a
single-column file still applies to every population. The .freq.gz that garlic writes, from
the analysis or from --freq-only, can be given straight back with --freq-file.

The automatic LOD cutoff identifies the two modes of the window LOD density by their sign.
The non-autozygous mode is the tallest peak below zero and the autozygous mode is the tallest

peak above zero, and the cutoff is the minimum between them. The previous rule could take two
peaks of the non-autozygous component (on the bundled chr21 example its cutoff was -55.7; it
is now -0.33). A density with no peak above zero gives no calls and a warning, with
lod_cutoff null in .params.json. A density with no peak below zero stops and asks for
--lod-cutoff. The substitution of 0 for a valley more than winsize from zero is gone. See
manual/lod-mode-rule.tex.

--gl-type documents that none of its forms suits low-coverage sequence data (below roughly
4x). See "Low-coverage sequence data" under Input Files.

Every run writes <out>.ind.tsv: per individual, the fraction of informative sites, the
median and 5th percentile of informative sites per window, and the observed/expected
heterozygote ratio in non-autozygous windows. Diagnostic only. See "The per-individual
information report" under Output Files.

--stratify-cutoff chooses the automatic LOD cutoff within strata of individuals that carry
the same evidence per window (grouped by the location of their non-autozygous LOD mode), so a
cohort of mixed depth is not over-called in its shallow members. A uniform cohort gets the
pooled cutoff exactly. Without the flag, garlic warns when the cohort would form more than
one stratum. See "Cohorts of mixed depth" under Input Files.

--exclude-regions <bed> drops the sites inside a mappability or repeat mask before anything is
computed, on every chromosome. This corrects the regional part of mapping bias, and it is the
only correction for mapping bias that garlic applies. See "Reference and mapping bias" under
Input Files.

24AUG2017 - v1.1.6 Modified behavior of --cm flag based on user feedback. When this flag is set ROH lengths and length classifications will be reported in cM but ROH boundaries will be reported in bps.

18JUL2017 - v1.1.5 Introduced --ld-subsample argument. If set > 0, will use only a subset of individuals for computing LD statistics when

calculating wLOD scores.

18MAY2017 - v1.1.4 If a freq file is provided, GARLIC now ensures that internal coding of alleles is consistent with the allele specified in the file. Improved handling of loci with 100% missing data. Improved reading of gzipped freq files. Improved lod score cutoff discovery. Removed default setting for --winsize, which must now be specified. Now allows --auto-winsize to be used with --winsize-multi, will choose best from the list.

12MAY2017 - v1.1.3 adds a new option for --gl-type called GQ, which is a phred-scaled probability that the genotype is wrong. If p = 0.999 is the probability that the genotype is correct, then GQ = -10*log10(1-p) = 30. The default for --gl-type is now set to none, so if you provide a TGLS file you will be required to choose between GQ, PL, and GL.

08MAY2017 - v1.1.2b fixes a bug that causes a crash when --overlap-frac is set to 0. Setting to zero now chooses the smallest sensible overlap fraction (1/winsize). If you wish to let garlic automatically guess overlap fraction, use the new command line option: --auto-overlap-frac. Also includes an interation counter to monitor progress of GMM portion of the code.

07MAY2017 - v1.1.2 fixes a bug that causes a crash when a tgls file indicates a genotype has 0 likelihood. Add progress bars for LD calculations and LOD score calculations. Improve efficiency of tgls file loading.

06MAY2017 - v1.1.1a fixes a bug introduced with 1.1.1 that caused crashed when --resample is used.

05MAY2017 - Update to version 1.1.1. For some large datasets (e.g. WGS), KDE can take extremely long to complete, even when
only a subset of individuals are considered. Here we introduce some changes to speed this calculation up. Instead of passing
all LOD score windows to the KDE function, we now pass only non-overlapping windows. This reduces the number of points sent
to the function by a factor of WINSIZE. If you wish to retain all LOD scores, you may set the --no-kde-thinning flag. We also change the default number of random individuals for KDE (--kde-subsample) to 20, which still may be set to 0 to use all individuals.
We also introduce a lower bound on the size of ROH reported. Now, an ROH will not be called unless it contains at least
WINSIZE*OVERLAP_FRAC number of SNPs. For example, for a WINSIZE (--winsize) of 100 and an OVERLAP_FRAC (--overlap-frac) of 0.25
the smallest possible ROH that will be reported will be 25 SNPs long.
Finally, if --overlap-frac is set to 0, GARLIC will attempt to guess a good choice.

02MAY2017 - Update to version 1.1.0a. Bug fixed that caused crashes when using TGLS file.

28APR2017 - Update to version 1.1.0. Includes several updates, including a method for reweighting LOD scores by gaps between SNPs, the option to report ROH lengths in cM, and the ability to provide per-genotype error rates.

25OCT2016 - Update to version 1.0.1. This update fixes a bug where ROH that extended upto the end of the chromosome failed to get assembled and reported. This update also introduces a new command-line flag --overlap-frac which is designed to reduce false positive calls by requiring a SNP be covered by at least OVERLAP_FRAC (default 0.25) proportion of high scoring windows to be included in an ROH call. This helps to reduce false positive calls near the boundaries of true ROH.

05MAY2016 - Initial release of GARLIC v1.0.0

A Command line reference

Generated from `garlic`'s own help strings by `make docs` in `src/`; do not edit `flags.tex` by hand.

Input files

`--freq-file` <string> A file of allele frequencies for every variant, with one frequency column per population headed by its label, or a single column that applies to every population: CHR SNP POS ALLELE FREQ <chr> <locus ID> <pos> <allele> <freq> By default the frequencies are calculated from the data. The .freq.gz that a run writes can be given back here.
Default: `none`

`--gl-type` <string> Which genotype-quality field to use. With `--vcf` it is read from the FORMAT column. With `--tgls` it names the form of the single value per genotype in the .tgls file. The two are not equivalent. With `--vcf` (recommended): GQ error = 10^(-GQ/10), as the VCF specification defines it. PL The whole PL array is converted to a posterior: P(g) proportional to 10^(-PL_g/10), error = 1 - P(called)/sum. A VCF normalises PL so that the called genotype's value is 0 for every call, and the information is in how much worse the alternatives are. GL The same, via PL = -10*GL. A missing field is an error that names whichever of GQ, PL and GL the file does carry, and '.' for a called genotype is an error. With `--tgls` (legacy, one value per genotype): GQ As above. The only form a VCF's values can be used in directly. PL error = 1 - 10^(-PL/10), where 10^(-PL/10) is P(genotype correct). This is not the VCF PL field. Integer PLs give errors of 0, 0.21, 0.37, 0.50, ..., and an error of 0.001 is PL = 0.00435. VCF PLs given here are rejected. Use `--vcf --gl-type` PL instead. GL error = 1 - 10^GL, with the same caveat. None of these forms suits low-coverage sequence data (below roughly 4x). Each gives one error rate per genotype, evaluated at the called genotype, so an uncertain homozygous call still counts as evidence for autozygosity. See "Low-coverage sequence data" in the manual.
Default: `none`

--map <string> A scaffold genetic map. Sites not in the file are interpolated, and sites outside its bounds are dropped. Required by **--weighted** and **--cm**. Format: <chr> <snpid> <genetic pos> <physical pos>. Negative positions are an error.
Default: `none`

--tfam <string> A TFAM file holding population and individual IDs.
Default: `none`

--tgls <string> A tgls file with one per-genotype value per individual, in the row order of the TPED. See **--gl-type** for what the values mean. For PL and GL this is not the VCF convention.
Default: `none`

--tped <string> A TPED file holding the map and the genotypes.
Default: `none`

--tped-missing <char> The one-character missing-data code in a TPED.
Default: `0`

--vcf <string> A VCF file (plain, gzipped or BGZF) to read genotypes from instead of **--tped**. Biallelic SNVs with a GT field are used. Multiallelic sites, indels and symbolic ALTs are skipped, and the counts are reported. Sample names come from the #CHROM line, so **--tfam** is not accepted. Use **--pop** for population labels. Sites must be grouped by chromosome. Ploidy must be 2, except on a chromosome declared with **--sex-system** or **--sex-chr**, where a haploid call is read as hemizygous. With **--phased** every call must use '|'.
Default: `none`

--vcf-pass-only <bool> Skip sites whose FILTER is neither PASS nor '.'. Without this flag such sites are kept and their number is reported, because FILTER records the variant caller's thresholds rather than garlic's.
Default: `false`

Populations

--pool-populations <bool> Analyse every individual as one population, ignoring the labels, as garlic did before per-population analysis. Output names are then <out>.roh.bed rather than <out>.<POP>.roh.bed. Allele frequencies are pooled over everyone, which inflates heterozygosity relative to any one population and biases the LOD scores. Use it when the labels are not populations (column 1 of a TFAM is the PLINK family ID) or to reproduce an earlier run.
Default: `false`

--pop <string> A file mapping sample ID to population, in place of the labels in the TFAM. Whitespace separated, '#' comments and blank lines ignored, .gz accepted: <sample_id> <population> [sex] The column order is the reverse of a TFAM's first two columns. The optional third column is sex in PLINK coding (1 male, 2 female, 0 or -9 unknown). Unlike in a TFAM, any other value is an error. Rows are matched by sample ID, not by order. Every sample in the data needs a row. Extra rows are ignored and counted, so one file can be used for a subset of the cohort.
Default: `none`

Genome and chromosomes

--autosomes-only <bool> Drop every chromosome that is not an autosome before calling: X, Y, Z, W and the mitochondrion by name, and 23, 24, 25 and 26 when `--build` names a human assembly. A hemizygous chromosome analysed as an autosome is called as one run along its whole length. Without a human `--build`, bare 23-26 are refused rather than dropped, because they are autosomes in a species with that many chromosomes. See `--sex-system`.
Default: `false`

--build <string> The genome build: hg18, hg19, hg38 or t2t-chm13 (T2T-CHM13v2.0, UCSC hs1). It supplies the centromere locations, the chromosome lengths and the human pseudoautosomal regions.
Default: `none`

--centromere <string> A file of custom centromere boundaries, <chr> <start> <end>. One of `--build`, `--centromere` or `--no-centromere` is required.
Default: `none`

--chr <string1> ... <stringN> Analyse only these chromosomes, e.g. `--chr` chr1 chr2 chrX. Names are matched after normalisation, so '1' and 'chr1' are the same. Naming a chromosome that is not in the data is an error.
Default: `all chromosomes in the input`

--chr-lengths <string> A file of chromosome lengths in two whitespace-separated columns, chromosome and length. A samtools faidx .fai is accepted as it is. Needed for `--froh-denominator` assembly on a non-human assembly or on TPED input, where nothing in the data states a length. Takes precedence over `--build` and a VCF's ##contig header.
Default: `none`

--exclude-regions <string> A BED file of regions whose sites are dropped before anything is computed, such as a mappability or repeat mask or segmental duplications. Mapping bias in such regions turns heterozygotes into homozygotes, which looks like autozygosity. Standard BED (0-based half-open <chrom> <chromStart> <chromEnd>, further columns ignored, track/browser/# lines skipped), gzipped or not. Chromosomes not in the data are ignored. The dropped sites are removed from the frequency file and from every window, but the FROH denominator is not reduced (unlike `--par`). Applied on every chromosome. Use `--par` for pseudoautosomal regions.
Default: `none`

--haploid-chr <string1> ... <stringN> Chromosomes haploid in every individual (mitochondrion, chloroplast). Dropped before calling.
Default: `names matching M or MT, plus 26 under a human --build`

--no-centromere <bool> Treat every chromosome as having no assembly gap, instead of requiring `--build` or `--centromere`. For assemblies with no centromere gap, and for references with no built-in table. Cannot be combined with `--build` or `--centromere`.
Default: `false`

--par <string1> ... <stringN> Pseudoautosomal regions on the shared sex chromosome, as <chr>:<start>-<end>, comma or space separated. The loci inside are dropped, since the regions are diploid in both sexes and too short to call. Any number of regions per chromosome is allowed. Coordinates are in the input's assembly, and a region cannot be named by chromosome code, because PLINK's 25 is an ordinary autosome in most species. `--build` supplies the human

X regions from the Genome Reference Consortium's definition (see centromeres/par_regions.txt). Naming any region here replaces them, and `--par` none turns them off.
Default: `the regions for --build, or none without one`

`--par-file` <string> The same regions read from a file, one per line as <chr> <start> <end>, with '#' comments and blank lines ignored and .gz accepted. Unlike `--centromere`, repeated rows for a chromosome add a region. May be combined with `--par`.
Default: `none`

`--sex-chr` <string1> ... <stringN> The chromosome that is diploid in the homogametic sex and hemizygous in the heterogametic one (X under xy, Z under zw). Needed only when it has another name, e.g. `--sex-system` zw `--sex-chr` LG12. Requires `--sex-system`.
Default: `the conventional name for --sex-system`

`--sex-chr-degenerate` <string1> ... <stringN> The chromosome carried only by the heterogametic sex (Y under xy, W under zw). It is dropped before calling, since no individual can carry a run of homozygosity on it.
Default: `the conventional name for --sex-system`

`--sex-system` <string> Which sex is heterogametic, and so which individuals are hemizygous on the shared sex chromosome: xy individuals coded 1 (male) in the TFAM or `--pop` are heterogametic zw individuals coded 2 (female) are heterogametic none every chromosome is diploid in every individual, and sex chromosomes are not reported PLINK codes 1 male and 2 female in every species, so in a ZW species the homogametic sex is coded 1. Leaving the flag unset is not the same as 'none'. If a sex chromosome is present and the flag is unset, garlic stops rather than analysing it as an autosome.
Default: `unset`

LOD scores and windows

`--M` <int> The expected number of meioses since a recent common ancestor, for `--weighted`.
Default: `7`

`--auto-winsize` <bool> Choose the window size automatically. With `--weighted` it is set from SNP density (see `--auto-winsize-coef`). Otherwise the search starts at `--winsize`, which must be given, and grows by `--auto-winsize-step` SNPs until the LOD score density meets `--auto-winsize-threshold`, up to `--max-winsize`.
Default: `false`

`--auto-winsize-coef` <double1> ... <doubleN> <slope> <intercept>: coefficients of winsize = slope*log(density) + intercept, used by `--auto-winsize` with `--weighted`. The defaults are an empirical fit to human SNP-array data. Change them for other ascertainment schemes or species.
Default: `8.3235 138.0521`

`--auto-winsize-step` <int> The step size, in SNPs, of the `--auto-winsize` search.
Default: `10`

`--auto-winsize-threshold` <double> The smoothness criterion at which the `--auto-winsize` search stops. Lower values demand a smoother LOD score density, and so a larger window.
Default: `0.5`

`--error` <double> The assumed genotyping error rate. Required.
Default: `-1`

--max-gap <int> No LOD score window is calculated across a gap between two loci larger than this many bp.
Default: `200000`

--max-winsize <int> The largest window size that `--auto-winsize` will try.
Default: `1000`

--mu <double> The mutation rate per bp per generation, for `--weighted`.
Default: `1e-09`

--phased <bool> The data are phased. Under `--weighted`, LD is then computed as r2 rather than hr2, which uses more memory. With `--vcf` every call must use '|'. Has no effect on the LOD scores without `--weighted`.
Default: `false`

--weighted <bool> Weight the LOD scores by LD and by the probabilities of no recombination and no mutation (wLOD). Requires `--map`.
Default: `false`

--winsize <int> The window size, in SNPs, over which LOD scores are calculated. With `--auto-winsize` it is where the search starts.
Default: `0`

--winsize-multi <int1> ... <intN> Several window sizes, in SNPs. garlic writes the LOD score KDE for each and calls no ROH. With `--auto-winsize` it chooses the best size from the list.
Default: `-1`

Calling runs of homozygosity

--auto-overlap-coef <double1> ... <doubleN> <slope> <intercept>: coefficients of overlap% = slope*log(density) + intercept, used by `--auto-overlap-frac`. The defaults are an empirical fit to human SNP-array data.
Default: `6.375 63.888`

--auto-overlap-frac <bool> Choose `--overlap-frac` from marker density (see `--auto-overlap-coef`).
Default: `false`

--het-rate-bounds <double1> ... <doubleN> <lo> <hi>: heterozygosity on the shared sex chromosome below which an individual reads as hemizygous and above which it reads as diploid. In between it is ambiguous, and garlic neither calls ROH there for that individual nor counts its alleles. Used to check each recorded sex and to infer the missing ones. The bounds are fixed rather than fitted, because a cohort of one sex has no bimodality to fit.
Default: `0.020000`

--ld-subsample <int> The number of individuals sampled at random for the LD calculations of `--weighted`. If the population has fewer, all are used. 0 or less uses every individual, which takes longer.
Default: `0`

--lod-cutoff <double> A LOD score cutoff above which windows are called autozygous, applied to every population. By default it is chosen automatically from the KDE of the LOD scores.
Default: `-999999`

--overlap-frac <double> The fraction of the windows covering a SNP that must be above the LOD cutoff for the SNP to be in a run. Runs are therefore at least WINSIZE*OVERLAP_FRAC SNPs long. If 0, garlic uses the smallest sensible value, 1/winsize.
Default: `0.25`

--sexchr-lod-cutoff <double> Call the shared sex chromosome at this LOD score cutoff instead of the autosomal one. Every run logs the cutoff that the sex chromosome's own windows would give, which is where a value for this comes from. The autosomal cutoff is the default because one chromosome's windows from part of a cohort often have no second minimum in the density.
Default: `the autosomal cutoff`

--sexchr-winsize <int> Use windows of this many SNPs on the shared sex chromosome instead of the autosomal size. The overlap fraction is unchanged. A LOD score cutoff is specific to the window size it was estimated at, so `--sexchr-lod-cutoff` is required as well.
Default: `the autosomal window size`

Choosing the LOD cutoff automatically

--kde-cut <double> Extend the density grid this many bandwidths beyond the range of the data.
Default: `3`

--kde-points <int> The number of grid points at which the LOD score density is evaluated. This also sets the resolution of the LOD cutoff.
Default: `512`

--kde-subsample <int> The number of individuals sampled at random for the LOD score KDE. If the population has fewer, all are used. The default, 0, uses every individual, so the cutoff depends on the data alone. A positive value saves memory but makes the cutoff depend on `--seed`.
Default: `0`

--kde-thin-step <int> Send every Nth LOD score window to the KDE. 0, the default, uses the window size, so the windows used do not overlap. 1 uses every window. `--no-kde-thinning` is a deprecated alias for 1.
Default: `0`

--mode-smooth-span <int> The number of adjacent grid points used to smooth the density when finding its modes. Must be smaller than `--kde-points`.
Default: `20`

--no-kde-thinning <bool> Deprecated. The same as `--kde-thin-step` 1, which sends every LOD score window to the KDE and can increase runtime considerably.
Default: `false`

--stratify-cutoff <bool> Choose the automatic LOD cutoff within strata of individuals that carry the same evidence per window, so that the shallow members of a cohort of mixed depth are not over-called. Individuals are sorted by the location of their non-autozygous LOD mode, neighbours whose modes differ by less than a quarter of the smaller spread share a stratum, and the pooled

rule is applied within each stratum. A cohort of uniform depth is one stratum and gets exactly the pooled cutoff. The per-individual mode, spread, stratum and cutoff are written to <out>.ind.tsv. Needs the automatic cutoff.
Default: `false`

Size classification

`--cm` <bool> Report ROH lengths and size classes in cM. ROH boundaries are still reported in bp. Requires `--map`.
Default: `false`

`--gmm-max-iter` <int> The maximum number of EM iterations when fitting ROH size classes.
Default: `1000`

`--gmm-tol` <double> The convergence tolerance of the ROH size-class EM fit.
Default: `1e-05`

`--nclust` <int> The number of Gaussian mixture components used to classify ROH lengths.
Default: `3`

`--size-bounds` <double1> ... <doubleN> The boundaries between ROH size classes, in increasing order. By default they are chosen automatically with a Gaussian mixture model (see `--nclust`).
Default: `-1.000000`

Counting classified genotypes

`--features` <string> A file classifying sites, in four whitespace-separated columns: <chr> <pos> <allele> <class> Lines beginning '#' are comments. Class labels are arbitrary text, and there may be any number of them. A site may appear more than once, so two annotation schemes can classify the same variant, but the same class twice at one site is an error. Chromosome names are matched ignoring case and a 'chr' prefix. For each individual garlic counts the classified genotypes inside runs of each size class and outside them, and writes <out>.counts.tsv. Genotypes are read from the input the run was called from, or from `--tped-counting` or `--vcf-counting` if given.
Default: `none`

`--roh-file` <string> Count classified genotypes against the calls in an existing garlic .roh.bed instead of calling ROH. `--tped` and `--vcf` are not used. Give the genotypes with `--tped-counting` and `--tfam-counting` or with `--vcf-counting`, and the classification with `--features`. A .roh.bed does not record which chromosomes were analysed or what the size class letters mean, so no site can be marked UNASSESSED and the size class boundaries are reported as unknown. Counting inside the run that makes the calls knows both. garlic 1.1.6a was released with two chromStart conventions under one version string, so the convention is inferred from the file and reported.
Default: `none`

`--tfam-counting` <string> The TFAM naming the samples in `--tped-counting`, in column order. Individuals are matched to the ROH calls by ID, and one present in only one of the two files is reported and left out of the table.
Default: `none`

`--tped-counting` <string> A TPED to count classified genotypes from, instead of the file the run was called from, for example when the calls come from array data and the classified variants are sequenced.

The two need not share a single site. Requires `--tfam-counting` and `--features`. These genotypes take no part in calling and are not filtered, since monomorphic and rare sites are what is being counted.
Default: `none`

`--vcf-counting` <string> A VCF to count classified genotypes from, instead of the file the run was called from. Sample names come from its #CHROM line. Requires `--features`. Unlike the VCF a run is called from, indels and multiallelic sites are read here. The classified allele is matched against REF and every ALT, so a classification that is not a biallelic SNV still counts.
Default: `none`

Output

`--force` <bool> Overwrite existing output files. Without it garlic will not overwrite an existing <out>.roh.bed.
Default: `false`

`--freq-only` <bool> Write the frequency file from the data and exit. Uses little memory.
Default: `false`

`--froh` <bool> Also write <out>.froh.tsv, with each individual's autozygous total and fraction by ROH size class. The denominator is stated in the header.
Default: `false`

`--froh-denominator` <string> What <out>.froh.tsv divides the autozygous total by. analyzed (default): the span the markers cover, from the first to the last locus of each analysed chromosome, less the centromere and any region excluded with `--par`. It moves with the marker set, so FROH from two different arrays are not directly comparable. assembly: the full length of each analysed chromosome, as published FROH denominators use. Lengths come from `--chr-lengths`, `--build` or a VCF's ##contig header, in that order, and a chromosome with no length stops the run. Not available with `--cm`.
Default: `analyzed`

`--out` <string> The base name for all output files.
Default: `outfile`

`--outdir` <string> Write all output files to this directory, which is created if needed. The same as prefixing `--out` with the path.

`--raw-lod` <bool> Also write the LOD scores to gzip-compressed files.
Default: `false`

Run control

`--dump-docs` <string> Write the command line reference to stdout and exit: txt for the README block, tex for the manual's appendix. Used by 'make docs'.
Default: `__none`

`--help` <bool> Prints this help dialog.
Default: `false`

--load-params <string> Read flag values from a <out>.params.json written by an earlier run. Flags on the command line take precedence, so a run can be repeated with one parameter changed.
Default: `__none`

--quiet <bool> Suppress the progress bar and informational messages. Errors and warnings still go to stderr and to <out>.error.
Default: `false`

--resample <int> The number of resamples for estimating allele frequencies. If 0, garlic uses the frequencies calculated from the data.
Default: `0`

--seed <int> The random seed. Affects `--kde-subsample`, `--ld-subsample` and `--resample`. If 0 (the default), a seed is drawn and written to the log so that the run can be repeated exactly.
Default: `0`

--threads <int> The number of threads for the LOD scores, the LD calculations under `--weighted`, the LOD score KDE and ROH assembly.
Default: `1`

--verbose <bool> Show the progress bar even when stderr is not a terminal. By default it is shown only on a terminal, because it is drawn with backspaces.
Default: `false`

--version <bool> Print the version and exit.
Default: `false`

-h <bool> Prints this help dialog.
Default: `false`

References

Cotter, D. J., Severson, A. L., Kang, J. T. L., Godrej, H. N., Carmi, S., and Rosenberg, N. A. 2024. Modeling the effects of consanguinity on autosomal and x-chromosomal runs of homozygosity and identity-by-descent sharing. *G3: Genes, Genomes, Genetics*, 14(2): jkad264.

Pemberton, T. J., Absher, D., Feldman, M. W., Myers, R. M., Rosenberg, N. A., and Li, J. Z. 2012. Genomic patterns of homozygosity in worldwide human populations. *American Journal of Human Genetics*, 91: 275–292.

Purcell, S., Neale, B., K., T.-B., Thomas, L., Ferreira, M. A. R., Bender, D., Maller, J., Sklar, P., de Bakker, P. I. W., Daly, M. J., and Sham, P. C. 2007. PLINK: a toolset for whole-genome association and population-based linkage analysis. *American Journal of Human Genetics*, 81: 559–575.